Brandon Gaona

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EDUCATION

Stevens Institute of Technology
B.S. Computer Science; B.S. Mathematics Hoboken, NJ

- **Coursework**: Discrete Math, Data Structures, Linear Algebra, Multivariable Calculus, Statistics, Computer Architecture, Algorithms, Differential Equations, Systems Programming
- **GPA**: 3.8 | **Awards**: PATHere Scholarship, Presidential Scholarship, Edwin A. Stevens Scholarship

EXPERIENCE

Federal Communications Commission – Office of Economics & Analytics (OEA) June – Aug. 2026
Machine Learning Intern Washington, DC

- Built and deployed an end-to-end Python machine learning pipeline (NLP, scikit-learn, FAISS, Flask) as sole developer, processing 373,000+ consumer complaints and achieving 93.7% classification accuracy while identifying 49,000+ duplicate or fraudulent submissions
- Reduced false-positive data quality flags from 79% to 7–8% and cut false fraud alerts from 150+ to 25 through signal recalibration; co-developed a Microsoft Copilot prompt engineering library adopted by FCC economists to improve report drafting efficiency

Hanlon Financial Systems Center Sept. 2025 – May 2026
SHIFT Research Assistant Hoboken, NJ

- Conduct quantitative research and backtesting of AMM ETF strategies across 10k+ historical data points per asset, modeling volatility, returns, and portfolio efficiency
- Performed statistical analysis and benchmarking against traditional ETFs, improving strategy evaluation accuracy and identifying performance differences of up to 15–25% in simulated returns

GDF Capital Inc May 2024 – Aug. 2025
Web Development Intern Doylestown, PA

- Developed company website using HTML, CSS, and JavaScript, supporting 100+ user interactions per session through dynamic content and form handling
- Optimized frontend performance and asset delivery, reducing page load times by 40%+ and improving overall user experience

PROJECTS

Consumer Complaint Intelligence Pipeline | *Python, NLP, scikit-learn, FAISS, spaCy, Flask*

- Designed and built a 5-layer, 15-stage machine learning pipeline for the Federal Communications Commission, using sentence embeddings, PCA, and clustering to organize 373,000+ consumer complaints into 25 categories and power an interactive dashboard
- Trained a classification model achieving 93.7% held-out accuracy and 0.938 macro-F1 score; engineered fraud detection identifying 49,000+ duplicate submissions (13%) and recalibrated anomaly detection to reduce false alerts from 150+ to 25

Custom 8-Bit CPU & Assembly Language

- Designed 8-bit single-cycle CPU with program counter, control unit, ALU, register file, and memory; defined minimal ISA (ADD, SUB, LOAD, STORE) with 2-bit opcode encoding
- Co-developed custom assembly language and Python assembler to compile source into machine code, successfully executing 20+ test programs in hardware simulation

Uncover – Fog-of-War Exploration Platform | *Next.js, React, TypeScript, Supabase*

- Built authenticated leaderboard API (Next.js) handling real-time ranking across 100+ users, with deterministic aggregation, tie-breaking logic, and time-windowed rankings backed by Supabase PostgreSQL
- Enforced row-level security and session-based auth; decoupled API response schema from database layer to enable independent frontend/backend iteration – Stevens QuackHacks 2026

Trading Backtest Dashboard | *FastAPI, PostgreSQL, Python, JavaScript, Docker*

- Developed full-stack backtesting platform executing trading strategies over multi-year datasets (1k+ trading days), returning performance metrics via REST API
- Engineered automated data ingestion with 100% idempotent upserts and optimized missing-data handling, eliminating redundant API calls and improving pipeline efficiency and reliability

SKILLS

Programming : Python, Java, C, C++, OCaml, R, HTML/CSS, Assembly, JavaScript, TypeScript, Bash
Frameworks : React, Next.js, Flask, Node.js, Microsoft Office Suite
Tools : Git, AWS, Supabase, Vercel, Jupyter, Linux
Concepts : Machine Learning, Quantitative Modeling, Backtesting, Statistical Analysis, Software Engineering, REST APIs, OAuth 2.0
Languages : Spanish (Fluent)